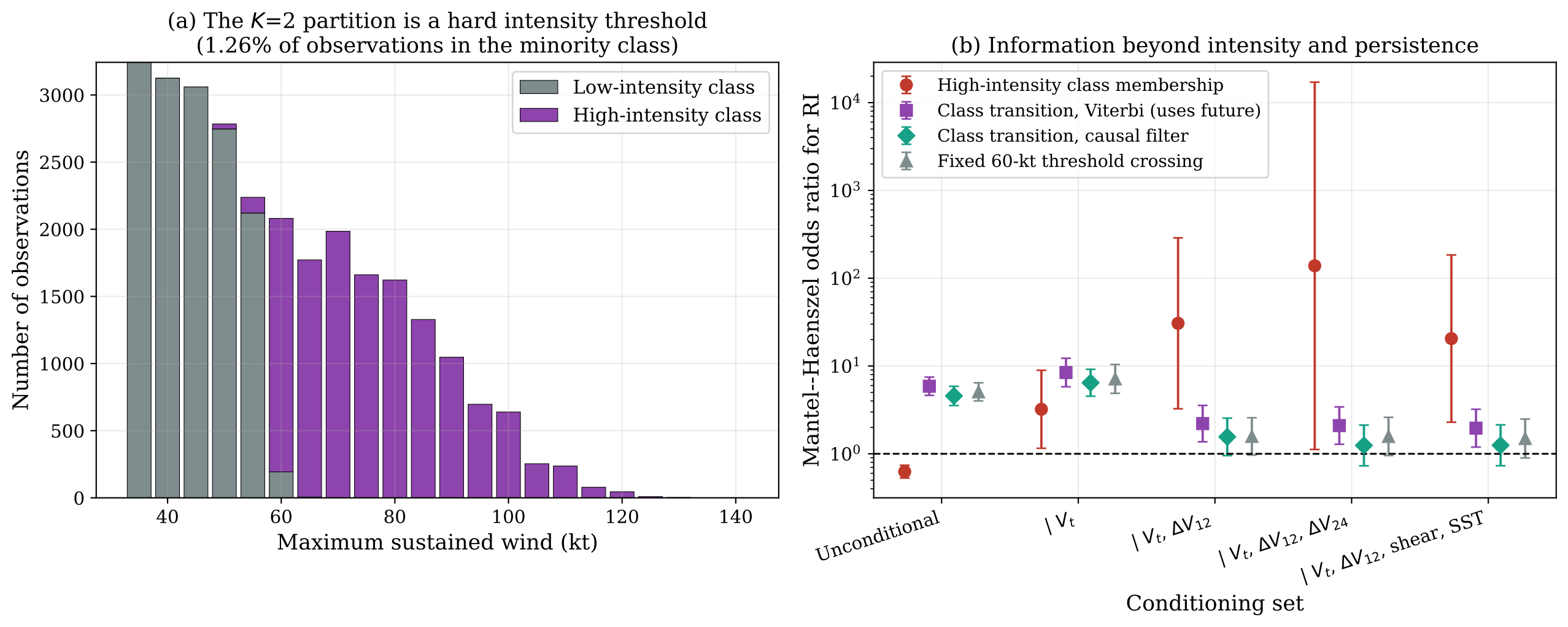
(a) The K=2 partition is a hard intensity threshold
(1.26% of observations in the minority class)
Low-intensity class
High-intensity class
Number of observations
Maximum sustained wind (kt)
0
500
1000
1500
2000
2500
3000
40
60
80
100
120
140
(b) Information beyond intensity and persistence
High-intensity class membership
Class transition, Viterbi (uses future)
Class transition, causal filter
Fixed 60-kt threshold crossing
Mantel--Haenszel odds ratio for RI
10^0
10^1
10^2
10^3
10^4
Unconditional
| V_t
| V_t, ΔV_{12}
| V_t, ΔV_{12}, ΔV_{24}
| V_t, ΔV_{12}, shear, SST
Conditioning set